CHAPTER B

Automated banking

Two questions per meeting, in 6 to 8 minutes correction included.
They do not constitute a list to be completed in full at each course.
This is based on the expected teaching, to be shifted if the concept has not yet been studied.

R001 · Fractions

Calculate $\frac{2}{3} + \frac{1}{4}$.

R002 · Priorities

Calculate $5 - 2(3 - 7)$.

R003 · Powers

Simplify $\dfrac{10^5 \times 10^{-3}}{10^4}$.

R004 · Roots

Simplify $\sqrt{50}$.

R005 · Proportion

18 represents 30% of a total. What total?

R006 · Direct rate

A price of 80 drops by 15%.

R007 · Initial value

After an increase of 20%, the value is 72.

R008 · Calculated rate

A quantity goes from 40 to 50.

R009 · Successive rates

20% increase then 20% decrease: overall rate?

R010 · Converse rate

What rate compensates for a 20% decrease?

R011 · Zero product

Solve (2 $x-1)(x+3)=0$.

R012 · Development

Development $(2x-3)^2$.

R013 · Factorization

Factor 3 x^2-12x.

R014 · Affine sign

Solve $-3x+6\geq 0$.

R015 · Slot

Slot per (1; 4) and (3; 0) ?

R016 · Right

Slope Equation -2 par (1; 4) ?

R017 · Weighted average

The values 2 and 5 are effective 3 and 2.

R018 · Median

Median of 1, 2, 2, 5, 8, 10?

R019 · Quartiles

For series 1, 2, 3, 4, 5, 6, 7, 8, give Q_1 and Q_3 with rows ceilings $\left\lceil \frac{n}{4} \right\rceil$ and $\left\lceil \frac{3n}{4} \right\rceil$.

R020 · Statistical standard deviation

For 1 and 3 equiponed, mean, variance and standard deviation?

R021 · Units

Convert 1 min 40 s to seconds; 2400 images over this duration give how much per second?

R022 · Simple probability

On a balanced dice, probability of a multiple of 3 ?

R023 · Complementary

$P(A) = \frac{3}{8}$. Provide $P(\overline{A})$.

R024 · Interval

Write $x \geq -2$ and $x <$ Three in the interval.

R025 · Logic

Denying "all the students have succeeded."

R026 · Histogram

Two classes of widths 2 and 4 have the same number 20. In a histogram with proportional areas what relative heights?

R027 · Circular diagram

A sector measures 72 °. Proportion?

R028 · Scale

An axis starts at 90. Bars stop at 100 and 110. Is the second value double the value of the first?

R029 · Percentage Percentage

40% of students make A; 25% make B. Proportion that makes A and B?

R030 · Proportion and strength

One category represents 0.125 of 80 people.

R031 · Defining and Representing a Sequence

Calculate the first four terms of $w_0 = 2,\ w_{n+1} = w_n + 2n + 1$; propose an explicit formula.

R032 · Arithmetic Sequences

We know. u_3 = 17 and u_8 = 37 for an arithmetic sequence. r and u_0.

R033 · Geometric sequences

Studying the variation in $u_n = -2 \times 3^n$ and then $v_n = 2(-1)^n$.

R034 · Amounts, indices and demonstrations

Calculate $\sum_{k=2}^{5} 2^k$ without summation symbol in the answer.

R035 · Conjecturing a limit and programming a threshold

For $u_n = 48(\frac{1}{2})^n$, determine the first rows where $u_n \leq 6$ and where $u_n < 6$.

R036 · Three forms, three readings

Give three forms of $g(x) = x^2 + 2x - 3$. Read its roots and minimum.

R037 · Discrimination and Resolution

Solve 3 $x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$.

R038 · Factoring without automating the discriminant

Determine the level 2 f polynomial such that $f(1) = f(4) = 0$ and $f(0) = -8$.

R039 · Sign, inequities and optimization

Solve $x^2 - 4x - 5 < 0$ then maximize $x(8 - x)$ on [0; 3].

R040 · Projection and angle

Two vectors have norms 2 and 5 and form 90 °, then 180 °. Calculate the products.

R041 · Coordinates, Norm and Orthogonality

For what real t the vectors (t; (2) and (3); -6) Are they orthogonal?

R042 · Vector Calculation and Identities

We give $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$.

R043 · Al-Kashi and choice of method

A triangle a $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine angle A.

R044 · The AB diameter circle

For $A(0; 0)$, $B(6; 0)$, determine M such as $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$.

R045 · Rate of change and secants

Calculate the rate of $f(x) = x^2 + 3x$ between 2 and $2 + h$.

R046 · Derivative number

For $f(x) = 3x^2 - 2x$, calculate $f'(1)$ from the rate.

R047 · Tangente and affine approximation

We give $f(3) = 5$ and $f'(3) = -2$. Write the tangent and approach $f(3{,}04)$.

R048 · When Differentiability Fails

Justify that $g(x) = |x - 2|$ is not derivable in 2. Also study the differentiability of $x \mapsto \sqrt{x}$ in 0 per its rate.

R049 · Conditioner: change denominator

Out of 60 objects, 20 are red; among the red, 5 are damaged. $P(R \cap D)$ and $P_R(D)$.

R050 · Trees, partitions and total probabilities

$P(A) = \frac{1}{3}$, $P_A(B) = \frac{3}{4}$, $P_{\overline{A}}(B) = \frac{1}{4}$. Calculate $P(B)$.

R051 · Independence and Incompatibility

We give $P(A) = 0{,}4$, $P(B) = 0{,}5$, $P(A \cap B) = 0{,}18$. Are they independent?

R052 · Successive Tests and Bernoulli

Four independent probability tests $p = \frac{1}{3}$. Calculation of the probability of four failures and then of less a success.

R053 · Simulate and discuss the model

A simulation of 2000 points gives 675 points under $y = x^2$. Which area does it estimate?

R054 · Derivative function and usual functions

Dirty x^5, x^{-2} and $\sqrt{x}$ specifying the areas.

R055 · Sum, product, inverse and quotient

Derivative $(x^2 + 1)(3x - 2)$ then $\frac{1}{x^2 + 1}$.

R056 · Sign of derivative and extrema

Study $f(x) = x^2 - 4x + 1$ on [0; 3] and give its extrema over this interval.

R057 · Parity, Inequalities and Comparison

Studying gender parity $x^4 + x^2$ and $x^3 - x$; Demonstrate $x^2 + 1 \geq 2x$.

R058 · Newton Optimization and Method

A perimeter rectangle 12 m wide x. Determine maximum range by bypass.

R059 · Normal Vector and Cartesian Equation

Equation of the right by (2; -1) normal (3; 4).

R060 · Orthogonal project and distance

Project P(4; 1) on $d : x = 2$.

R061 · Equation of a circle

Recognize $x^2 + y^2 + 2x - 8y + 17 = 0$.

R062 · Solve a configuration

Intersect ($x - 1)^2 + (y + 2)^2 = 9$ with $y = 1$.

R063 · Variable, Event and Law

Two balanced pieces are launched. X counts batteries. Write his law and $P(X \leq 1)$.

R064 · Expectation and linearity

Gross gain equals 10 with probability $\frac{1}{5}$ and 0 otherwise. What bet makes the game fair?

R065 · Variance, standard deviation and König·Huygens

For X values $-1, 1$ equiprobable, calculate expectation, variance and standard deviation.

R066 · Transform and Compare Earnings

$E(X) = 3$, $V(X) = 4$. Give the characteristics of $Y = -2X + 5$.

R067 · Samples and mean observed

For $\mu = 5$, $\sigma = 3$ and n=36, give the experimental interval requested.

R068 · Set a new function

What is the equation of tangent to e^x One?

R069 · Algebraic properties and comparisons

Simplify $\frac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$.

R070 · Dériver et étude

Dirty $g(t) = 3e^{-2t}$ and study its sense of variation.

R071 · From discreet to continuous

For $Q(t) = 5e^{0,3t}$, express $Q(n)$ and $\frac{Q(t+2)}{Q(t)}$.

R072 · Digitally build and demonstrate inequality

Show that $e^{2x} \geq 1 + 2x$ for all real x.

R073 · Measuring an angle in radians

Convert 45 °, 120° ; give the radius arc 2 m and angle $\frac{3\pi}{4}$.

R074 · Rewinding and image of a real

Place images of $\frac{7\pi}{2}$ and $-\frac{5\pi}{3}$ Bringing them back to a turn.

R075 · Sine, cosine and remarkable values

If cos $x = \frac{3}{5}$ and image of x is in the fourth quadrant, giving sin x.

R076 · Associated angles and symmetries

Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $7\frac{\pi}{4}$.

R077 · Approximate pi and reinvest

For n=6 and c=1, calculate the new string after doubling.

R078 · Graphical reading

In Figure R078, read $f(1)$, resolve $f(x) = 0$ and $f(x) > 0$.

